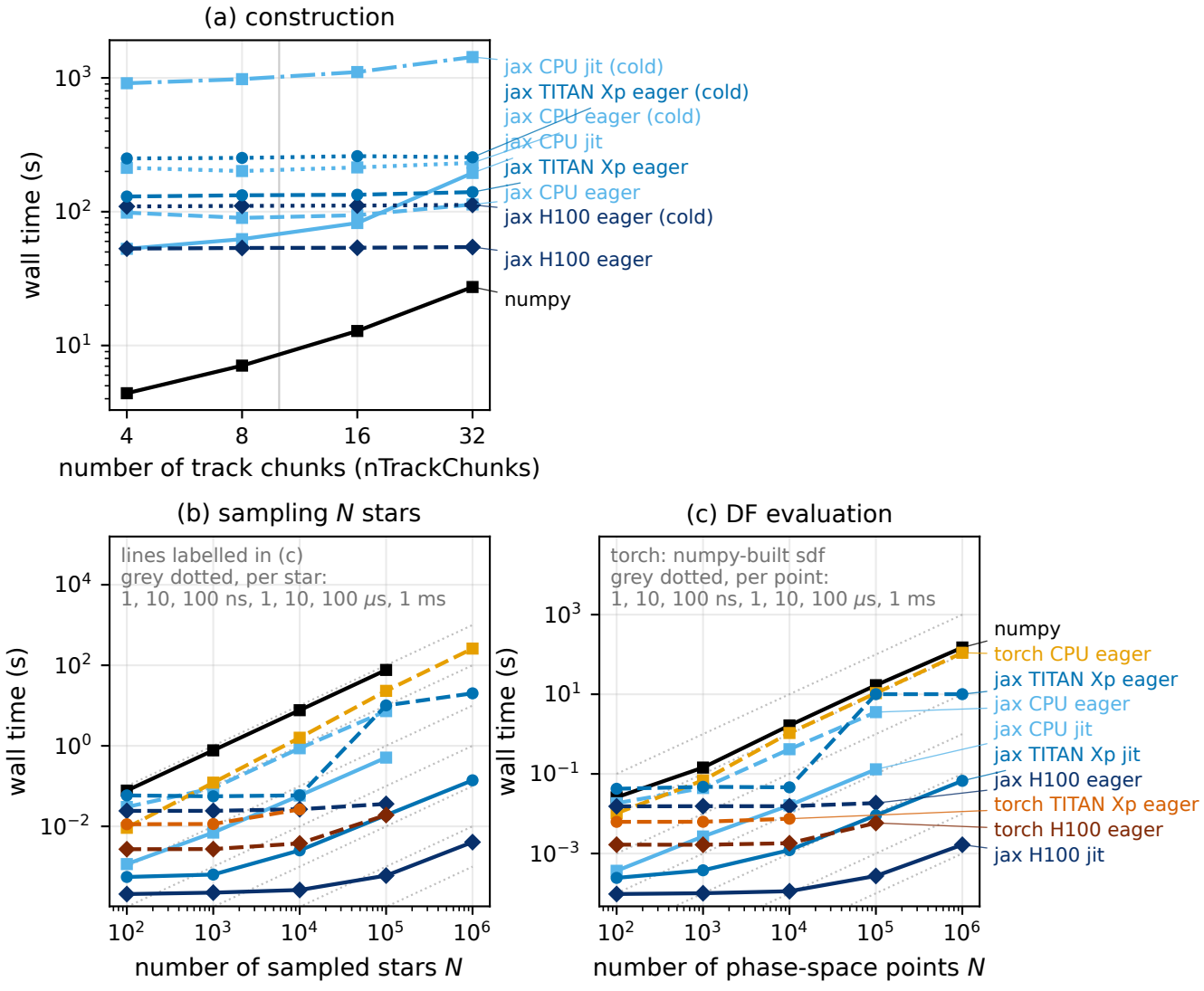
(a) construction
wall time (s)
number of track chunks (nTrackChunks)
jax CPU jit (cold)
jax TITAN Xp eager (cold)
jax CPU eager (cold)
jax CPU jit
jax TITAN Xp eager
jax CPU eager
jax H100 eager (cold)
jax H100 eager
numpy
(b) sampling N stars
lines labelled in (c)
grey dotted, per star:
1, 10, 100 ns, 1, 10, 100 μs, 1 ms
wall time (s)
number of sampled stars N
(c) DF evaluation
torch: numpy-built sdf
grey dotted, per point:
1, 10, 100 ns, 1, 10, 100 μs, 1 ms
wall time (s)
number of phase-space points N
numpy
torch CPU eager
jax TITAN Xp eager
jax CPU eager
jax CPU jit
jax TITAN Xp jit
jax H100 eager
torch TITAN Xp eager
torch H100 eager
jax H100 jit